Smart Zakat

Software Requirements Specification

Version 2.0

Status: Draft

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Revision History

Name	Date	Reason For Changes	Version	Date of Approval
[Author's Name]	22 Oct 2026	Initial draft based on high-fidelity website screens 1–10	1.0	Pending

1. Introduction

1.1 Purpose

This document specifies the software requirements for **Smart Zakat, Release 1.0**, a web application (smartzakat.org) that lets donors calculate Zakat, distribute it among verified recipients, pay through local payment methods, and receive proof of how every taka was used. The SRS covers the complete public website and donor-facing system as shown in the ten high-fidelity screens (Landing, Login/Register, Donor Dashboard, Zakat Calculator, Donate, General Donation Campaigns, Verified Patient Cases, Distribute & Pay, Proof of Payment, History & Auto-Pay). Back-office tools (hospital onboarding, scholar-board review, admin moderation) are referenced where they affect donor-facing behaviour but are not specified in detail.

1.2 Document Conventions

For tracking changes, mark them in the document with color (changes, additions) or cross them out (deletions). Requirement and item IDs use the prefixes S- (stakeholder), U- (user class), UC- (use case), BR- (business rule) and NFR- (non-functional requirement). "Shall" denotes a mandatory requirement. "TBD" means to be detailed (see Appendix D). Currency is Bangladeshi Taka (BDT).

1.3 Intended Audience and Reading Suggestions

ID	Stakeholder	Description
S-1	Customer / Product Owner	Checking correspondence of business goals and functionality requirements to the expectations from implementing the product.
S-2	Development team	Forming the accurate vision of the project, detailed functional and nonfunctional requirements.
S-3	QA team	Making test-plans and test-cases.
S-4	PM/BA	Estimating the quote of the project, planning resources and the timeline of work.
S-5	User (Donor)	Basing on this document the Terms of Service and the Privacy Policy are created.

ID	Stakeholder	Description
S-6	Scholar board / Shariah advisors	Validating Zakat calculation rules, eligibility rules and recipient categories.
S-7	Partner hospitals and charities	Understanding how cases, bills and campaigns are displayed, funded and paid.

Reading suggestion. Everyone should start with Sections 1 and 2. Developers and QA then read Sections 3, 5 and 6 and Appendix C. Customers and scholar board focus on Sections 2, 5 (business rules) and Appendix D. Section 4 and Appendix B give the data view.

1.4 Product Scope

Smart Zakat is a new, self-contained product that makes Zakat and charitable giving simple, transparent and traceable. It helps donors to:

- calculate Zakat in minutes with an easy calculator;
- give Zakat to eligible recipients, or give general donations (Sadaqah) to hospital and charity campaigns;
- fund verified, anonymised patient bills directly;
- receive proof of payment (receipts with bill references, privacy protected);
- automate yearly giving (for example every Ramadan) and review lifetime history.

Objectives: build donor trust through verification and proof; increase giving through low-friction payment (bKash/Nagad, card, bank transfer); direct funds to medical need. Current showcased impact figures on the landing page are BDT 2.4M donated, 120+ patients helped and 15 partner hospitals. A separate Vision and Scope document is not yet available (see Appendix D).

1.5 References

#	Document	Version / Location
R-1	Smart Zakat Website High-Fidelity Screens (10 screens)	Smart_Zakat_Website_High_Fidelity.pdf
R-2	SRS Template	SRS_Template_docx.pdf
R-3	bKash, Nagad, card gateway and bank transfer API documentation	TBD – provider portals
R-4	Applicable Bangladesh data protection, financial and charity regulations	TBD

2. Overall Description

This section gives the general view of the product, its environment, expected users, constraints, assumptions and dependencies.

2.1 Product Perspective

Smart Zakat is a new, self-contained web product. It integrates with external payment providers, SMS/e-mail gateways and (indirectly) partner hospital billing records. It is not a replacement for an existing system. Hospital and charity data (patient cases, bills, campaigns) is supplied by partners and verified before publication; Zakat eligibility of recipients is confirmed by a scholar board.

2.2 Product Functions

ID	Function	Screen
F-1	Present product value, impact statistics and how-it-works flow	1
F-2	Register and sign in by e-mail/phone and password, or phone OTP; password recovery	2
F-3	Donor dashboard: Zakat due, total given, recent impact, auto-pay and recent donations	3
F-4	Zakat calculator (cash, gold, silver, investments, debts; nisab check; 2.5% rate)	4
F-5	Choose giving mode: Pay Zakat, General Donation, Fund a Patient's Bill	5
F-6	Browse and filter campaigns; donate preset/custom amount, anonymously, optionally monthly	6
F-7	Browse verified, anonymised patient cases and fund a bill	7
F-8	Distribute Zakat among recipients (manual or auto-distribute), choose payment method, pay, set up auto-pay	8
F-9	Proof of payment: confirmation, PDF receipt, impact breakdown, share impact	9
F-10	Giving history, statistics, auto-pay management, proof viewing, yearly Zakat statement	10

Major groups: **Identity** (F-2), **Calculation** (F-4), **Giving** (F-5–F-8), **Transparency** (F-1, F-9), **Account** (F-3, F-10).

2.3 User Classes and Characteristics

ID	User class	Description
U-1	Guest (not signed in)	Visitor who can view the landing page, use the calculator, view campaigns and cases, but must sign in to pay. Limited rights.

ID	User class	Description
U-2	Donor (signed-in user)	Registered user who has signed in. Can calculate, pay Zakat/donations, set auto-pay, view dashboard, receipts and history.
U-3	Administrator	Back-office staff who manage campaigns, cases, recipients, percentages and content. (Out of detailed scope.)
U-4	Scholar board member	Reviews recipient categories and confirms Zakat eligibility. (Out of detailed scope.)
U-5	Partner hospital / charity representative	Supplies verified cases, bills and payment confirmations. (Out of detailed scope.)

Donors are expected to be general public users with mixed technical skill, mostly on mobile phones; the interface must therefore be simple and clear.

2.4 Operating Environment

- Responsive web application running in current versions of Chrome, Safari, Firefox and Edge (desktop and mobile browsers).
- Server side hosted on a cloud or local data centre (TBD) with a relational database.
- Interacts with payment gateways (bKash, Nagad, card acquirer, bank transfer), SMS gateway (OTP) and e-mail service.
- Vision and Scope document with further detail is not yet available.

2.5 Design and Implementation Constraints

- **Religious/business rules:** Zakat is due when net wealth stays above nisab for a full lunar year, at 2.5%; only scholar-approved recipient categories may receive Zakat.
- **Labelling:** every gift shall be tracked and labelled either Zakat or Donation; general donations are never counted as Zakat.
- **Privacy:** patient identities shall be anonymised (e.g. "P-***41"); no personal medical identifiers may be displayed.
- **Payments:** use only licensed payment providers; no card data stored by Smart Zakat (PCI DSS scope reduction).
- **Language and currency:** BDT; English UI in release 1.0 (Bangla TBD, see Appendix D).
- **Standards:** HTTPS/TLS only; WCAG 2.1 AA targeted; JSON/REST for integrations.
- Technology stack, hosting and database are not specified by the customer (TBD; see 2.7).

2.6 User Documentation

- Online help / FAQ (including "About Nisab" explanation).
- Terms of Service and Privacy Policy.
- Short in-app guidance for each step of How it works (Calculate > Choose recipients > Pay > Get proof).
- Administrator manual for back-office users (format TBD).

2.7 Assumptions and Dependencies

Assumptions

- A-1: Nisab thresholds (gold/silver based) and current metal prices are provided by an admin-maintained setting or external price feed; the customer has not specified which (TBD).
- A-2: Zakat rate is fixed at 2.5% of net zakatable wealth.
- A-3: Default Zakat distribution is 40% patient bills, 35% surgeries, 25% charity medicine, 0% charity food, adjustable by the donor and by admin.
- A-4: Partner hospitals confirm bill payments so receipts can reference real bill numbers.
- A-5: Technology stack and hosting not given; the development team will propose them, following the client's interests.

Dependencies

- D-1: Availability and API contracts of bKash, Nagad, card and bank-transfer providers.
- D-2: SMS gateway for phone OTP.
- D-3: Timely delivery of verified cases and campaigns by partner hospitals.
- D-4: Scholar-board approval of eligibility rules before launch.

3. External Interface Requirements

This section defines the tools, software and database elements with which the system interacts.

3.1 User Interfaces

The website is made of ten high-fidelity screens (Appendix C and reference R-1). Common standards:

- Global header on all screens: logo "Smart Zakat" and links Home, Calculator, Donate, Cases, History, Sign In.
- Responsive layout (desktop and mobile); primary actions as prominent buttons (e.g. Calculate Zakat, Donate Now, Pay).
- Amounts always shown as "BDT n,nnn" with thousands separators.
- Every gift is visibly labelled "Zakat" or "Donation".
- Error messages appear inline under the field, in plain language; the Screen Entry Exception Tables in Section 5 define field constraints.
- Verification badges ("Verified partner", "Verified by hospital", "Reviewed by scholar board") are shown wherever relevant.

Detailed UI design is documented in the separate high-fidelity design (R-1), not in this SRS.

3.2 Hardware Interfaces

No direct hardware interfaces. Supported devices are standard desktops, laptops, tablets and smartphones with a modern browser. Receipts may be downloaded as PDF files to the device storage.

3.3 Software Interfaces

Interface	Purpose	Notes
bKash / Nagad payment APIs	Mobile-wallet payments and recurring auto-pay	Versions TBD
Card payment gateway	Card payments	Tokenised; no card storage
Bank transfer	Bank payment / confirmation	Reconciliation method TBD
SMS gateway	Phone OTP, payment notifications	
E-mail service	Receipts, reminders, password reset	
Relational database	Users, cases, campaigns, donations, receipts	
PDF generation library	Receipts and yearly Zakat statements	
Gold/silver price source	Nisab and valuation of metals	TBD

3.4 Communications Interfaces

- HTTPS (TLS 1.2+) for all browser–server communication; REST/JSON APIs internally and to providers.
- Webhooks/callbacks from payment providers for payment status; the system shall verify signatures.
- E-mail (SMTP/API) and SMS for notifications.
- Sharing of impact via standard web share links (no personal data in shared content).

4. Domain Model

Main entities, attributes and relationships:

Entity	Key attributes	Relationships
User (Donor)	id, name, email, phone, password hash, status	has many Donations, AutoPay plans, Calculations
ZakatCalculation	id, cash, gold_g, silver_g, investments, debts, zakatable_wealth, net_wealth, zakat_due, date	belongs to User
Recipient	id, name, type (Hospital / Charity), category (Patient bills, Surgeries, Medicine, Food), default_percent, scholar_approved	has many Allocations
Campaign	id, title, category (Surgery/Dialysis/Medicine), goal, raised, percent_funded, verified_partner	belongs to Recipient

Entity	Key attributes	Relationships
PatientCase	id (e.g. A-2291), title, category, amount_needed, percent_funded, hospital_verified, anonymised_patient_id	belongs to Hospital Recipient
Donation	id, user, date, type (Zakat / Donation), amount, anonymous, recurring, payment_method, status	has many Allocations, one Receipt
Allocation	id, donation, recipient, bill/case/campaign ref, amount, percent	links Donation to Recipient/Case/Campaign
Receipt	id (e.g. ZK-10482), donation, pdf, issued_at	belongs to Donation
AutoPay	id, user, amount, schedule (e.g. Ramadan yearly), method, next_run, status	belongs to User

See Appendix B for the relationship overview.

5. System Features (Use Cases)

5.1 Use Case UC-1 Register / Sign In

Brief Description	Allows a user to create an account or sign in with e-mail/phone and password, or with a phone OTP, to manage their giving.
Business Trigger	The user clicks Sign In, or attempts to pay while not signed in.
Preconditions	The user is on smartzakat.org; the user is not signed in.

Basic Flow — Assumption: the user already has an account.

Line	System Actor Action	System Response
1	The user opens the Sign In page.	The system displays the Welcome back form (Email or phone, Password, Sign In, Forgot password?, Continue with phone OTP, Create an account).
2	The user enters email/phone and password and clicks Sign In.	The system validates the credentials.
3	–	The system opens the Donor Dashboard (UC-8) or returns the user to the page they came from.

Post Condition: The user is authenticated and a session is created.

Alternate Flow (AF1): Sign in with phone OTP. If at line 1 the user clicks "Continue with phone OTP":

Line	System Actor Action	System Response
1	The user enters a phone number.	The system sends a one-time code by SMS.
2	The user enters the code.	The system validates the code and signs the user in.

The use case restarts at line 3 of the Basic Flow.

Alternate Flow (AF2): Create an account. If at line 1 the user clicks "New here? Create an account":

Line	System Actor Action	System Response
1	The user enters name, email/phone and password.	The system validates the data and sends a verification code.
2	The user confirms the code.	The system creates the account and signs the user in.

The use case restarts at line 3 of the Basic Flow.

Alternate Flow (AF3): Invalid credentials. If at line 2 the credentials are wrong, the system shows "Incorrect email/phone or password" and keeps the user on the form. The use case restarts at line 1 of the Basic Flow. After 5 failed attempts the account is temporarily locked (BR-1).

Alternate Flow (AF4): Forgot password. If at line 1 the user clicks "Forgot password?", the system sends a reset link/code to the registered e-mail/phone and lets the user set a new password. The use case terminates.

Business Rules:

1. BR-1: Lock sign-in for 15 minutes after 5 consecutive failed attempts.
2. BR-2: Email and phone shall each be unique per account.

Non Functional Requirements: OTP valid for 5 minutes; passwords hashed with a modern algorithm (bcrypt/argon2).

Data Requirements: User (email/phone, password hash, status), OTP (code, expiry, attempts).

Prototype Screen: Screen 2 – Login / Register (smartzakat.org/login).

Screen Entry Exception Table:

#	Field	Constraint	Response
1	Email or phone	Valid email format or valid Bangladesh mobile number	Message: "Enter a valid email or phone number"
2	Password	Min. 8 characters on registration	Message: "Password must be at least 8 characters"

5.2 Use Case UC-2 Calculate Zakat

Brief Description	Lets a user enter their assets and liabilities and see whether Zakat is due and the amount.
Business Trigger	The user clicks Calculate Zakat / Calculator.
Preconditions	None (guests may calculate; signing in is only needed to pay and save).

Basic Flow — Assumption: the user enters valid numeric values.

Line	System Actor Action	System Response
1	The user opens the Zakat Calculator.	The system displays fields: Cash & bank savings, Gold (grams), Silver (grams), Investments / business stock, Debts & liabilities (deduct), and the "About Nisab" note.
2	The user enters or edits asset values.	The system recalculates and updates "Your result" live: Zakatable wealth, Debts deducted, Net wealth.
3	–	If net wealth is at or above nisab, the system shows "Your total is above nisab, so Zakat applies" and Zakat due (2.5%).
4	The user clicks Continue to Donate.	The system carries the Zakat amount to Distribute & Pay (UC-4).

Post Condition: A calculation record exists (saved to the account if signed in).

Alternate Flow (AF1): Below nisab. If at line 3 net wealth is below nisab, the system displays "Your wealth is below nisab; Zakat is not due" with Zakat due BDT 0 and disables Continue to Donate (offering General Donation instead). The use case terminates.

Alternate Flow (AF2): Guest continues. If at line 4 the user is not signed in, the system redirects to UC-1 and returns to the result afterwards.

Business Rules:

1. BR-3: Net wealth = Zakatable wealth − Debts and liabilities.
2. BR-4: Zakat due = 2.5% of net wealth when net wealth ≥ nisab, otherwise 0.
3. BR-5: Gold and silver weights are converted to BDT using the current admin-maintained or feed price (TBD).
4. BR-6: The lunar-year holding condition is explained to the user; the user confirms it (TBD whether tracked automatically).

Non Functional Requirements: Result updates in under 1 second; amounts rounded to the nearest taka.

Data Requirements: ZakatCalculation entity (Section 4).

Prototype Screen: Screen 4 – Zakat Calculator.

Screen Entry Exception Table:

#	Field	Constraint	Response
1	Cash & bank savings, Investments, Debts	Non-negative number, max 12 digits	Message: "Enter a valid amount"
2	Gold / Silver (grams)	Non-negative number, up to 2 decimals	Message: "Enter weight in grams"

5.3 Use Case UC-3 Choose How to Give

Brief Description	Lets the donor pick one of three giving modes: Pay Zakat, General Donation, or Fund a Patient's Bill.
Business Trigger	The user clicks Donate / Donate Now.
Preconditions	None.

Basic Flow

Line	System Actor Action	System Response
1	The user opens Donate.	The system shows three cards: Pay Zakat (Zakat), General Donation (Donation / Sadaqah), Fund a Patient's Bill (Zakat or Donation).
2	The user selects a card (Start / Browse campaigns / View cases).	The system opens the Calculator (UC-2), Campaigns (UC-5) or Verified Patient Cases (UC-6) respectively.

Post Condition: The chosen giving flow is opened.

Business Rules: BR-7: Every gift shall be labelled Zakat or Donation from this point on.

Prototype Screen: Screen 5 – Donate: Choose How to Give.

5.4 Use Case UC-4 Distribute and Pay Zakat

Brief Description	Lets a signed-in donor split their Zakat among recipients, choose a payment method, pay, and optionally make the payment automatic.
Business Trigger	The user clicks Continue to Donate or Pay Now.

Preconditions	The user is signed in; a Zakat amount is available.

Basic Flow — Assumption: auto-distribute is on and the payment succeeds.

Line	System Actor Action	System Response
1	The user opens Distribute and pay.	The system shows the amount (editable), the "Auto-distribute fairly" switch, recipients with category and percentage (City Hospital 40%, Mercy Medical Centre 35%, Care Foundation 25%, Orphan Aid Trust 0%) and the payment methods.
2	The user keeps the auto-distribution (or adjusts percentages).	The system recalculates the Summary (e.g. City Hospital BDT 5,000; Mercy Medical BDT 4,375; Care Foundation BDT 3,125; Total BDT 12,500).
3	The user selects a payment method (bKash/Nagad, Card, Bank transfer).	The system shows the payment details required by that method.
4	The user clicks "Pay BDT 12,500".	The system redirects/opens the provider payment step and processes the payment.
5	–	On success the system records the donation and allocations, issues a receipt and opens the Proof of Payment (UC-7).

Post Condition: The donation is recorded as type Zakat; allocations and receipt exist; recipients are notified.

Alternate Flow (AF1): Make this automatic. If at line 4 the user enables "Make this automatic" and selects the repeat interval (e.g. Ramadan, yearly), the system saves an AutoPay plan (amount, schedule, method) before payment. The use case continues at line 4.

Alternate Flow (AF2): Manual percentages do not total 100%. If at line 2 the percentages do not sum to 100%, the system shows an error and disables Pay. The use case restarts at line 2.

Alternate Flow (AF3): Payment fails or is cancelled. If at line 4 the payment fails, the system shows the reason, keeps the distribution and offers retry or another method. The use case restarts at line 3. No receipt is issued.

Business Rules:

1. BR-8: Allocation percentages shall total exactly 100%; amounts are rounded to whole taka with the remainder assigned to the largest share.
2. BR-9: Only recipients with scholar-approved eligibility are listed for Zakat ("Reviewed by scholar board").
3. BR-10: Auto-distribute uses the default percentages configured by admin.

Non Functional Requirements: Payment confirmation shown within 10 seconds of provider callback.

Data Requirements: Donation, Allocation, AutoPay, Receipt.

Prototype Screen: Screen 8 – Distribute & Pay.

Screen Entry Exception Table:

#	Field	Constraint	Response
1	Amount (BDT)	Positive integer, min. BDT 100 (TBD)	Message: "Enter an amount of at least 100"
2	Recipient percentage	0–100, integer	Message: "Percentages must total 100%"

5.5 Use Case UC-5 Make a General Donation to a Campaign

Brief Description	Lets a donor browse verified campaigns and donate (Sadaqah) a chosen amount, optionally anonymously or monthly.
Business Trigger	The user chooses Browse campaigns.
Preconditions	Active campaigns exist; the user is signed in to complete payment.

Basic Flow

Line	System Actor Action	System Response
1	The user opens Active campaigns.	The system lists campaigns (e.g. Emergency Surgery Fund 72%, Free Dialysis Centre 45%, Child Medicine Pack 60%) with the "Verified partner" badge and progress.
2	The user filters by All / Surgery / Dialysis / Medicine.	The system shows only matching campaigns.
3	The user clicks Select on a campaign.	The system marks it and shows "Your donation".
4	The user picks 500, 1,000, 2,500 or Other and enters the amount.	The system updates the donation amount.
5	The user clicks Donate Now.	The system proceeds to payment (as in UC-4 lines 3–5) with type Donation.

Post Condition: A Donation (not Zakat) is recorded with a receipt.

Alternate Flow (AF1): Donate anonymously. If at line 4 "Donate anonymously" is ticked, the donor name is hidden from the campaign/partner. The use case continues at line 5.

Alternate Flow (AF2): Repeat monthly. If at line 4 "Repeat monthly" is ticked, the system creates a monthly recurring donation plan. The use case continues at line 5.

Business Rules: BR-11: General donations are not counted as Zakat in dashboards, history or the yearly Zakat statement.

Prototype Screen: Screen 6 – General Donation: Campaigns.

Screen Entry Exception Table:

#	Field	Constraint	Response
1	Amount (Other)	Positive integer, min. BDT 50 (TBD)	Message: "Enter a valid amount"

5.6 Use Case UC-6 Fund a Verified Patient Case

Brief Description	Lets a donor pay all or part of a verified, anonymised hospital bill directly.
Business Trigger	The user chooses "Fund a Patient's Bill" or opens Cases.
Preconditions	At least one verified case is open; the user is signed in to pay.

Basic Flow

Line	System Actor Action	System Response
1	The user opens Verified patient cases.	The system lists cases with title, bill ID, amount needed, "Verified by hospital", % funded and note "Patient identity is protected" (e.g. Heart surgery #A-2291, BDT 90,000, 65%).
2	The user filters by category.	The system shows matching cases.
3	The user clicks "Fund this bill".	The system asks for amount and giving type (Zakat or Donation).
4	The user confirms amount and type and pays.	The system processes payment (as UC-4 lines 3–5) and allocates funds to the hospital bill.

Post Condition: The case funded percentage is updated; the receipt references the bill number.

Alternate Flow (AF1): Case fully funded. If at line 3 the case reaches 100%, the system disables funding and shows "Fully funded". The use case terminates.

Alternate Flow (AF2): Amount exceeds remaining need. If at line 3 the amount is higher than the remaining need, the system limits it to the remaining amount or offers the surplus to another case (TBD).

Business Rules: BR-12: Zakat may only fund cases whose recipients are scholar-approved as eligible. BR-13: Only hospital-verified cases are displayed; patient identity is never shown.

Prototype Screen: Screen 7 – Verified Patient Cases.

5.7 Use Case UC-7 View Proof of Payment

Brief Description	Shows the donor that a payment succeeded and where the money went, with a downloadable receipt.
Business Trigger	A payment completes, or the user clicks View proof.
Preconditions	A completed donation exists belonging to the user.

Basic Flow

Line	System Actor Action	System Response
1	–	The system shows "Donation successful" with amount and date, and the receipt page (smartzakat.org/receipt/ZK-xxxxx).
2	The user reviews "Where your money went".	The system lists each allocation with recipient, bill/pack reference, amount and anonymised patient ID (e.g. City Hospital – Bill #A-2291, BDT 5,000, Patient P-***41).
3	The user clicks Download Receipt (PDF).	The system generates and downloads the PDF.

Alternate Flow (AF1): Share impact. If at line 2 the user clicks "Share impact", the system offers share options with a message containing no personal or patient data. The use case terminates.

Post Condition: The receipt is available in History.

Business Rules: BR-14: Receipts show the note "Zakat eligibility verified by scholars" only for Zakat gifts. BR-15: Only the owner may view a receipt.

Prototype Screen: Screen 9 – Proof of Payment.

5.8 Use Case UC-8 View Dashboard, History and Manage Auto-Pay

Brief Description	Gives the donor an overview of Zakat due, giving totals, impact, recent and past donations, and lets them manage auto-pay and download a yearly Zakat statement.
Business Trigger	The user signs in or opens History.
Preconditions	The user is signed in.

Basic Flow

Line	System Actor Action	System Response
1	The user opens the Dashboard.	The system shows Zakat due this year (e.g. BDT 12,500) with Pay Now, Total given and This year, recent impact, auto-pay card and recent donations; sidebar: Overview, Zakat, Donations, Auto-pay, Receipts, Settings.
2	The user opens History.	The system shows Lifetime giving, This year, People helped, the Auto-pay plan, and the Past donations table (Date, Recipient, Type, Amount, Proof).
3	The user clicks View proof on a row.	The system opens the receipt (UC-7).
4	The user clicks "Download yearly Zakat statement".	The system generates a PDF of Zakat-type donations for the year.

Alternate Flow (AF1): Manage auto-pay. If at line 1 or 2 the user clicks Manage, the system lets them change amount, method or schedule, pause or cancel the plan. The use case terminates.

Alternate Flow (AF2): No donations yet. If at line 2 no donations exist, the system shows an empty state with a link to Calculate Zakat.

Business Rules: BR-16: Yearly statement includes only Zakat-type gifts. BR-17: Auto-pay shall notify the donor at least 3 days before each run and the donor may cancel before it executes.

Prototype Screens: Screen 3 – Donor Dashboard; Screen 10 – History & Auto-Pay.

5.9 Use Case UC-9 View Landing Page

Brief Description	Presents the product, impact figures and entry points to calculate or donate.

Business Trigger	A visitor opens smartzakat.org.
Preconditions	None.

Basic Flow

Line	System Actor Action	System Response
1	The visitor opens the home page.	The system shows the hero ("Give your Zakat with confidence"), buttons Calculate Zakat and Donate Now, impact figures (BDT 2.4M donated, 120+ patients helped, 15 partner hospitals), three feature highlights and the How it works flow.
2	The visitor clicks Calculate Zakat or Donate Now.	The system opens UC-2 or UC-3.

Business Rules: BR-18: Impact figures are computed from recorded data and refreshed at least daily.

Prototype Screen: Screen 1 – Landing Page.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- NFR-1: Pages load in ≤ 3 seconds on a 4G connection for 95% of requests.
- NFR-2: Calculator results update in ≤ 1 second.
- NFR-3: Support at least 1,000 concurrent users, with peaks during Ramadan (scalable to 5,000).
- NFR-4: PDF receipt generation in ≤ 5 seconds.

6.2 Safety Requirements

- Patient privacy: no real names, photos or medical identifiers shall be shown; only anonymised IDs.
- Financial safety: payment shall be atomic; a donation is recorded only when the provider confirms payment; duplicate submissions are prevented (idempotency keys).
- Auto-pay protection: donors are notified before each charge and may cancel; failed auto-pay never retries more than 3 times.
- Funds shall be disbursed only to verified recipients; discrepancies are flagged for admin review.

6.3 Security Requirements

- Authentication by password or phone OTP; passwords hashed; sessions expire after inactivity (30 min).
- Role-based access control (Guest, Donor, Admin, Scholar, Partner).
- All traffic over HTTPS; sensitive data encrypted at rest; no card data stored.
- Protection against OWASP Top 10 (injection, XSS, CSRF); rate limiting on login and OTP.
- Audit log of payments, allocations and admin changes.

- Compliance with applicable Bangladesh data protection and financial regulations (TBD).

6.4 Software Quality Attributes

Attribute	Requirement
Usability (preferred over ease of learning)	A first-time donor completes Calculate ⟶ Pay in ≤ 5 minutes without help; mobile-first; WCAG 2.1 AA.
Availability	99.5% monthly uptime; 99.9% during Ramadan.
Reliability	No lost or duplicated transactions; daily reconciliation with providers.
Maintainability	Modular code, documented APIs, automated tests ≥ 70% coverage.
Portability	Latest two versions of Chrome, Safari, Firefox and Edge; Android and iOS browsers.
Interoperability	Payment providers swappable through an adapter layer.
Correctness	Calculator results verified against scholar-approved test cases.
Testability	Staging environment with provider sandboxes.
Adaptability	Rates, nisab, percentages and recipient lists configurable without code changes.

7. Other Requirements

- **Database:** relational database with daily backups, 7-year retention of financial records (TBD with legal).
- **Internationalization:** English in release 1.0; Bangla planned; currency BDT only; dates shown as DD Mon YYYY.
- **Legal:** Terms of Service, Privacy Policy and charity/payment regulatory compliance to be confirmed.
- **Reuse:** payment, receipt and notification modules designed for reuse in future mobile apps.
- **Reporting:** admin reports on donations, allocations, recipient payouts and campaign progress (TBD).

Appendix A: Glossary

Term	Description
Zakat	Obligatory annual charity (2.5% of net zakatable wealth above nisab).

Term	Description
Sadaqah	Voluntary charity; labelled "Donation" in the system.
Nisab	Minimum wealth threshold above which Zakat is due after a full lunar year.
Zakatable wealth	Sum of cash, gold, silver and investments/business stock.
Net wealth	Zakatable wealth minus debts and liabilities.
Campaign	A hospital or charity fundraising programme (e.g. Free Dialysis Centre).
Case	A verified, anonymised patient bill that can be funded directly.
Auto-pay	Scheduled recurring payment (e.g. yearly at Ramadan).
Distribute	Splitting a payment among several recipients by percentage.
bKash / Nagad	Mobile financial services in Bangladesh.
OTP	One-time password sent by SMS.
BDT	Bangladeshi Taka.
TBD	To be detailed.

Appendix B: Analysis Models

Entity relationship overview

User 1—* ZakatCalculation; User 1—* Donation; User 1—* AutoPay; Donation 1—* Allocation; Donation 1—1 Receipt; Allocation *—1 Recipient; Allocation *—0..1 PatientCase; Allocation —0..1 Campaign; Recipient 1— Campaign; Recipient 1*— PatientCase.

Giving flow (state)

Calculate → Choose recipients → Pay → (Payment pending → Succeeded / Failed) → Receipt issued → Visible in History.

Formal diagrams (class, data flow, state machine) are to be added in the next revision.

Appendix C: Interface Examples

Screen 1 – Landing Page (smartzakat.org). Header navigation; hero text; buttons Calculate Zakat, Donate Now; statistics (BDT 2.4M, 120+, 15); three feature cards (Easy Zakat calculator, Verified patient cases, Proof of payment); How it works strip.

Screen 2 – Login / Register.

Field	Description
Email or phone	Editable text field. Valid email or Bangladesh mobile number.
Password	Editable masked field. Min. 8 characters.
Forgot password?	Link to reset flow.
Sign In	Button; submits credentials.
Continue with phone OTP	Button; starts OTP flow.
Create an account	Link to registration.

Screen 3 – Donor Dashboard. Sidebar (Overview, Zakat, Donations, Auto-pay, Receipts, Settings); greeting "Assalamu Alaikum, <name>"; cards: Zakat due this year with Pay Now, Total given and This year; Your recent impact (hospital, bill ID, amount); Auto-pay card with Manage; Recent donations list (date, recipient, type, amount).

Screen 4 – Zakat Calculator.

Field	Description
Cash & bank savings	Editable number (BDT).
Gold (grams)	Editable number (g).
Silver (grams)	Editable number (g).
Investments / business stock	Editable number (BDT).
Debts & liabilities (deduct)	Editable number (BDT).
Zakatable wealth, Debts deducted, Net wealth	Not editable, auto-calculated.
Zakat due (2.5%)	Not editable, auto-calculated.
Continue to Donate	Button; enabled only when Zakat is due.

Screen 5 – Donate: Choose How to Give. Three cards: Pay Zakat (Start), General Donation (Browse campaigns), Fund a Patient's Bill (View cases), each with a Zakat/Donation label.

Screen 6 – General Donation: Campaigns.

Field	Description
Category filter	Tabs: All, Surgery, Dialysis, Medicine.
Campaign card	Title, "Verified partner", % of goal progress bar, Select button.
Amount (BDT)	Quick buttons 500 / 1,000 / 2,500 and Other (editable number).
Donate anonymously	Checkbox.

Field	Description
Repeat monthly	Checkbox.
Donate Now	Button. Note: general donations are not counted as Zakat.

Screen 7 – Verified Patient Cases. Filter tabs; case cards with title and bill ID, Needed amount, "Verified by hospital", % funded, "Fund this bill" button, "Patient identity is protected" note.

Screen 8 – Distribute & Pay.

Field	Description
Amount (BDT)	Editable number; pre-filled from calculator.
Auto-distribute fairly	Toggle.
Recipients	Name, category, percentage (editable when auto-distribute is off).
Payment method	Selector: bKash / Nagad, Card, Bank transfer.
Summary	Not editable; per-recipient amounts and total.
Make this automatic / Repeat every	Toggle and dropdown (e.g. Ramadan yearly).
Reviewed by scholar board	Informational badge.
Pay BDT n	Button.

Screen 9 – Proof of Payment. Success status, amount and date, Download Receipt (PDF), Share impact, "Where your money went" list (recipient, bill ref, amount, anonymised patient ID), note "Zakat eligibility verified by scholars."

Screen 10 – History & Auto-Pay. Statistic cards (Lifetime giving, This year, People helped); Auto-pay summary; Past donations table (Date, Recipient, Type, Amount, Proof – View proof link); "Download yearly Zakat statement" link.

Appendix D: To Be Determined List

1. Calculator inconsistency in Screen 4: cash BDT 450,000 + investments BDT 120,000 = 570,000, yet Zakatable wealth shows BDT 530,000 and net wealth BDT 500,000; gold (40 g) valuation and the intended formula must be clarified.
2. • The Dashboard Total given (BDT 87,500) differs from the sum of the four History rows (BDT 92,500); clarify how the lifetime total is computed.
3. Nisab basis (gold or silver) and the source of metal prices.
4. Whether the lunar-year holding period is tracked by the system or confirmed by the user.
5. Source and approval process of the default distribution percentages and scholar-board eligibility rules; can Zakat fund a general campaign?
6. Minimum donation amounts and fee handling (does the donor or the platform bear payment fees?).

7. Technology stack, hosting, and exact payment-provider API versions.
8. Languages (Bangla), legal/regulatory requirements, and data retention.
9. How surplus funds are handled when a case is over-funded.
10. Back-office (admin, scholar, partner) screens and reports; a Vision and Scope document.
11. Wording: Screen 9 says "Donation successful" for a Zakat payment; confirm the label.